Appendix II. Debt Sustainability Analysis

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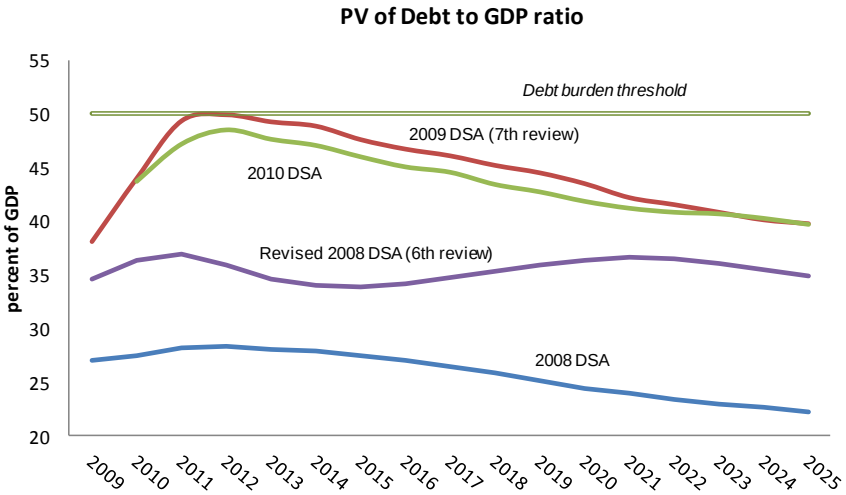
The debt sustainability outlook for Cape Verde is broadly unchanged compared to the 2009 DSA, but debt levels have increased significantly from the 2008 DSA*. The temporary acceleration of external borrowing for investment to address growth and export impeding infrastructure bottlenecks increased debt trajectories relative to projections in the 2008 DSA. As a result, the PV of external debt-to-GDP ratio approaches the risk threshold in 2012 under the baseline and breaches it in some stress tests. However, most of the debt is highly concessional and on terms longer than 10 years, maintaining the debt service ratios far below their thresholds. To restore buffers against future shocks, Cape Verde will need to (i) reduce external borrowing from 2012 onwards, (ii) mobilize domestic revenue, and (iii) keep domestic borrowing low to accumulate foreign reserves and reinforce the credibility of the currency peg.*



Source : IMF staff projections. Baseline

I. BACKGROUND

1. **This report updates the debt sustainability analysis (DSA) prepared in 2009 (IMF Country Report No. 09/328).** The baseline scenario reflects the latest IMF discussions with the authorities (September 2010). Since the 2008 DSA, Cape Verde has been adversely affected by the global crisis through lower exports (mainly services) and FDI. These developments reduced GDP growth and contributed to a weakening of the balance of payments and the fiscal accounts. However, economic growth shows clear signs of a recovery in 2010 and is expected to strengthen over the medium term.

2. **The DSA is based on end-2009 data provided by Cape Verde's authorities.** The debt data currently covers central government external debt, publically-guaranteed debt, and domestic debt.

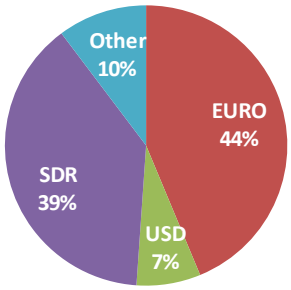
3. **Public debt is mostly external and highly concessional**, and the recent increase in the stock is driven by both multilateral and bilateral creditors. The World Bank Group and the African Development Banks are the major external creditors (more than 60 percent of the total outstanding debt). The debt is mainly denominated in euros and SDRs.

Text Table 1. Cape Verde: Stock of Public Debt, 2008-2009

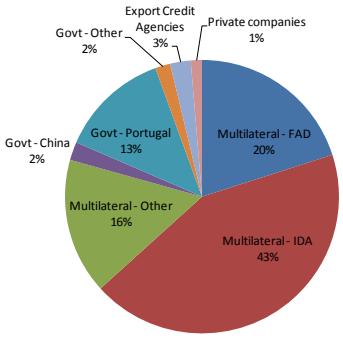
	Million of US $		Percent of total		Percent of GDP	
	2008	2009	2008	2009	2008	2009
Total	1006.2	1154.0	100	100	65.0	63.9
External	671.9	774.6	66.8	67.1	47.3	47.3
Domestic	334.3	379.3	33.2	32.9	17.7	16.5

Source: Cape Verde's authorities; and IMF and World Bank staff estimates.

Cape Verde: Stock of Debt by Currency, 2009



Cape Verde: Composition of Public External Debt, 2009



4. **The 2009 increase in external debt in percent of GDP ends a five-year decline, from 63 percent in 2003.** The substantial rise in 2009 is due to a higher fiscal deficit, but also to the planned acceleration of foreign-financed public investment, in response to the international crisis. However, the 2009 outturn for in public investment was slightly lower than scheduled in the 2008 DSA, as some projects were delayed. Nominal external debt at the end of 2008 and 2009 was 47.3 percent of GDP, with an average maturity of 25 years. A downward revision of historical nominal GDP numbers has mechanically increased the debt ratio in percent of GDP. The appreciation of the dollar in 2009 was unfavorable to Cape Verde, given its peg to the euro, increasing the ratio of foreign debt to GDP. However, authorities have recently canceled or postponed the implementation of a few non-priority

projects, in order to curb debt accumulation. The share of commercial external debt is only 1 percent.

5. **Public debt data does not include a share of the debt of state owned enterprises (SOEs).** SOE debt includes loans contracted through the government and already taken into account in the DSA framework. However, SOEs also have arrears to providers and credit lines from banks that are included in contingent liabilities in the risks to the debt outlook. The authorities' report on contingent liabilities to the Council of Ministers in August 2009 estimates the total liabilities owed by the airline, TACV, and the water and electricity company, Electra (both SOEs) at 13 percent of GDP (although the estimated amount needed to recapitalize these firms would be lower, at around 5 percent of GDP).

6. **Domestic debt is kept low, in order to protect the peg to the euro.** Domestic debt represents one third of total debt and is composed mostly of long-term bonds owed to domestic banks and the public pension fund. It has declined significantly since 2005 as a result of the strong fiscal consolidation under the PSI, which is anchored on keeping domestic borrowing low to build external reserves.

II. THE DSA BASELINE SCENARIO

7. **Relative to the previous DSA, the macroeconomic framework incorporates the impact of the crisis and the policy response, but remains broadly unchanged regarding the medium and long-term perspectives.** More specifically, real GDP growth is expected to rise over the medium term and catch-up to its pre-crisis average, reflecting increased private investment, structural reform and the world recovery. However, the aftermath of the international financial crisis has led to a slight downward revision of real growth projections in the short term. After the global crisis abates, external demand and FDI should recover, supported by the new infrastructure. Growth will be driven by tourism, FDI, and to a lesser extent, remittances. Under the baseline, the growth rate is assumed to recover slowly. The current account balance has been revised up, reflecting weaker domestic demand in the short term and a higher export capacity in the medium term, as a consequence of the accelerated public investment program.

Text Table 2. Cape Verde : Key Macroeconomic Assumptions, 2010-30

	2010-11	2012-30
Real GDP growth (percent)		
Updated	5.7	5.3
Previous [1]	5.9	5.3
Fiscal Revenue (percent of GDP)[2]		
Updated	23.3	25.3
Previous [1]	23.9	25.2
External Current Account (percent of GDP)		
Updated	-17.5	-7.5
Previous [1]	-22.6	-7.9

Sources : Cape Verde's authorities; and IMF and world Bank staff estimates.

[1] Previous DSA covers the period 2009-29

[2] Total revenues excluding grants

Box 1. Macroeconomic Assumptions for the Baseline Scenario

Real GDP growth slowed due to the global crisis, but is expected to recover over the medium term. Despite the acceleration of infrastructure spending and the lower tax rates, the slowdown in FDI could keep growth below its historical average for some time. The acceleration of public infrastructure spending is unlikely to boost contemporaneous growth because of the small fiscal multipliers caused by high import leakages and because it is not fully offsetting the demand effect of the decline in private investment. While the global shock worsened the near-term growth outlook, Cape Verde's medium-term growth prospects remain strong.

The acceleration of the public investment program decided in 2009 as a policy response to the crisis is reflected in the short-term growth rate of the construction sector. However, the second order impact of public investment, such as higher competitiveness or positive externalities in the private sector, are not taken into account in the baseline medium and long term growth assumptions.

Inflation is assumed to remain subdued. Given the high import component of domestic demand and the peg to the euro, GDP deflator and consumer price inflation are assumed to be low in the medium and long term.

Current account balance: Imports are assumed to remain broadly flat as a share of GDP. Imports of intermediate goods for tourism are assumed to grow, but this will be largely offset by a decline in other imports. The deceleration of FDI is holding back imports in the near term, and completion of the current large investment projects will contain imports of capital goods in the long run: construction material, equipment, and labor are mostly imported. In addition, as Cape Verde gets wealthier, emigration will slow and the share of remittances and other current transfers in GDP should slowly decline.

Government balances: After a temporary shortfall due to lower activity and tax rate reductions as a policy response to the crisis, tax revenues are assumed to gradually increase and stabilize at around 24 percent of GDP. As the country climbs the income ladder, the informal sector should shrink and the individual and corporate income tax are assumed to increase slightly to reach 8 percent of GDP. At the same time, as a consequence of Economic Partnership Agreements and other Free Trade Agreements (FTA), international trade taxes are decreasing to 4.5 percent of GDP, from 5.5 percent before the crisis. Grants are assumed to gradually decline to 1 percent of GDP. After a peak above 40 percent in the short term on account of accelerated public investment, total expenditures should stabilize around 33 percent, with a wage bill at 12 percent and capital spending at 11 percent. The overall balance (including grants) should fall from a deficit of 15 percent of GDP in 2010 and stabilize at a 2 percent deficit in the medium term. The primary basic balance (excluding grants, interest, and foreign-financed expenditure) is expected to be slightly positive (on average) in the long run.

External assistance: Cape Verde will progressively borrow on less concessional terms. The baseline scenario assumes that the concessionality of new loans will decline in the medium term. With Cape Verde's graduation from least-developed-country status in 2008, it acquired "blend" status with multilateral development agencies, notably the EIB, ADB, and the IBRD. It is now eligible to borrow from their non-concessional windows, and, after a transition period, access to concessional funds will shrink.

Domestic borrowing: The baseline assumes that the authorities will continue to keep domestic borrowing low to support accumulation of foreign reserves and protect the exchange rate peg. Fiscal policy is geared toward accumulation of reserves to 4.5 months of current-year imports in the medium term, and stabilization thereafter.

The assumption is maintained that fiscal revenue (including grants) will remain relatively flat in the medium and long term, with a substantial shortfall in the short term, reflecting weaker activity and tax cuts in response to the global economic turmoil. Tax revenues are expected to increase slightly in the long run, and will balance the probable decline in external grants. New public borrowing (both domestic and external) is assumed to be moderate over the medium term, reflecting the progressive closing of the concessional external financing window, as well as the commitment to protect the stock of reserves.

III. EXTERNAL DEBT SUSTAINABILITY

Baseline scenario

8. **The LIC debt sustainability framework is guided by country-specific debt burden thresholds for external debt, based on the strength of a country's policies and institutions.** Cape Verde now ranks as a 'top performer' under the joint IMF/World Bank debt sustainability framework, based on its three-year moving average CPIA score. The indicative external debt burden thresholds for countries in this category are a present value (PV) of debt-to-exports ratio of 200 percent, a PV of debt-to-revenue ratio of 300 percent, a PV of debt-to-GDP ratio of 50 percent, and debt-service-to-exports and revenues ratios of 25 and 35 percent, respectively.

9. **Cape Verde's external debt appears sustainable.** Under the baseline scenario, all debt indicators remain below their thresholds through 2030 (Text Table 3). The pace of external borrowing will slow sharply from 2012, putting the PV of the external debt-to-GDP ratio on a declining path thereafter. This policy commitment is embedded in the MTFF (Medium Term Fiscal Framework) submitted to the National Assembly with the 2010 budget proposal, and in the revised MTFF submitted to staff in September 2010. While the debt-to-GDP ratio will nearly reach the debt burden threshold in 2012, all other debt ratios will remain well below their thresholds, including the debt service ratios. Furthermore, Cape Verde benefits from a large and steady flow of remittances that can increase the repayment capacity of the country. Remittances have averaged 12 percent of GDP over the past ten years and the projections conservatively assume an average of 8 percent of GDP over the next 10 years.

Text Table 3. Cape Verde: Baseline Debt Ratios, 2010-30

	Threshold	2010	Medium Term 2010-15	Long Term 2016-30
External				
PV of debt-to-GDP	50.0	43.7	46.9	40.5
PV of debt-to-exports	200.0	121.2	119.1	73.8
PV of debt-to-revenue	300.0	189.6	198.1	152.7
Debt service-to-exports	25.0	5.0	4.5	5.7
Debt service-to-revenue	35.0	7.8	7.5	12.0
Fiscal				
PV of debt-to-GDP		61.7	64.6	50.9
PV of debt-to-revenue		204.9	227.5	178.4
Debt service-to-revenue		17.9	19.9	27.3

Sources : Cape Verde's authorities; and IMF and world Bank staff estimates.

Alternative scenario and stress tests

10. **Alternative scenarios and bound tests show that all debt indicators except the PV of debt-to-GDP remain well below their thresholds through 2030.** The breach of the PV of debt-to-GDP threshold under the growth shock (B1) is minor (the deviation only amounts to 1 percent of GDP) and lasts for only two years. Other shocks have a more sustained impact. For instance, the export shock (B2) and US deflation shock (B3) result in breaches of 6 percent of GDP in 2012 and last several years. Under the exchange rate shock (B6), which assumes a 30 percent devaluation of the escudo, the PV of the debt-to-GDP ratio breaches the threshold during several years.

11. **The other debt ratios provide assurances that Cape Verde's repayment capacity remains strong.** Given the high export-to-GDP ratio (over 40 percent) and the strong capacity to collect taxes (the tax collection rate is far above regional standards), the debt-to-export and debt-to-revenue ratios remain well below the thresholds even under extreme scenarios. Moreover, all of Cape Verde's external debt has maturities that are typically longer than 10 years, which alleviates cash flow pressures and limits rollover difficulties. This lessens concerns about the country's ability to repay its debt.

12. **A high-growth and a low-growth scenario were constructed to test the sensitivity of debt ratios to growth assumptions.** Each scenario assumes that growth will deviate from the baseline by +/–1.0 percentage points in 2011–16. As in the baseline, the low-growth scenario implies that nominal GDP will be permanently below its pre-crisis trend (the 2011–16 deviations in growth rate are permanent shocks to the GDP level). Because the high-growth scenario assumes growth slightly above its pre-crisis average, GDP will eventually surpass the trend.

13. **During the low-growth scenario, the debt service dynamics remain sustainable.** The PV of external debt-to-GDP remains below its threshold, with a peak in 2012. The ratios decline slowly thereafter, ending at 36 percent of GDP in 2029, compared with 30 percent of GDP under the baseline scenario. Under the low-growth scenario, the borrowing space would be very limited in the near-term. Given the assumption that borrowing will continue at long maturities, the debt ratios will stay low even in the low-growth scenario. This should dispel any concerns about the country's ability to repay its debt.

14. **Under the high growth scenario, with a higher growth dividend than conservatively assumed under the baseline, the debt stock ratios would decline rapidly after 2012.** If the higher growth dividend materializes, the ensuing strong tax revenues will help contain the deficit, reduce debt, and further expand public expenditures.

IV. PUBLIC SECTOR DEBT SUSTAINABILITY

15. **The DSA baseline shows that public debt sustainability will be preserved, since the trajectory of total public debt mirrors external debt and yields the same conclusion (Table 4).** The PV of total debt-to-GDP peaks at 67 percent of GDP in 2012 and declines thereafter in line with the authorities' commitment in the MTFF to reduce external borrowing from 2012. External borrowing is determining the trajectory of total public debt because net domestic debt (gross domestic debt net of deposits in the BCV and commercial banks) is projected to be kept below 20 percent of GDP. The debt-to-revenue and debt service-to-revenue ratios will remain broadly stable.

16. **Alternative scenarios and bound tests indicate that debt sustainability indicators remain on paths similar to the external public debt.** Only the scenario of an unchanged primary balance from 2010 leads to an unsustainable situation, but it is not relevant since 2010 is an exceptional year, characterized by fiscal stimulus and an accelerated public investment program.

17. **Other risks to the debt outlook include a currency mismatch and contingent liabilities of state-owned enterprises.** A depreciation of the euro could raise the debt ratios. The current cost to recapitalize Electra and TACV is estimated at 5 percent of GDP.

V. CONCLUSIONS: DEBT SUSTAINABILITY BROADLY UNCHANGED

18. **The debt sustainability outlook is broadly similar to the 2009 DSA.** However, compared to the 2008 DSA, the external and public-sector debt sustainability analysis reveals increased vulnerability as external debt ratios are higher. This reflects an acceleration of public investment, a depreciation of the euro (and thus the CVE) compared to the dollar, and the technical downward revision of the historical nominal GDP numbers.

19. **Staff assesses Cape Verde's debt as sustainable.** The PV of debt-to-GDP ratio breaches its threshold under half of the stress tests. However, Cape Verde's strong policies and institutions should help ensure that the growth dividends from the accelerated public investments materialize, and help restore buffers in the medium term. Moreover, the long maturities and high concessionality of the existing debt ensure that liquidity risks are low, as shown by debt service indicators.

20. **Still, increased debt vulnerabilities call for caution.** In particular:

- The current debt assessment is based on several assumptions, including (i) the authorities' commitment to decelerate external borrowing in 2012 and thereafter; (ii) the authorities' capacity to raise revenue in the medium term; (iii) the authorities' ability to contain expenditures; (iv) a rebound in economic growth.
- New borrowing should continue to be mostly at long maturities and on concessional terms. Long maturities will ensure low debt service obligations relative to Cape Verde's export and revenue ratios. If maturities are shorter, and concessionality lower compared to the baseline, the risks to the debt outlook would worsen.

Table 1.: External Debt Sustainability Framework, Baseline Scenario, 2007-2030 1/
(In percent of GDP, unless otherwise indicated)

	Actual			Historical 0 Average	Standard 0 Deviation	Projections									
	2007	2008	2009			2010	2011	2012	2013	2014	2015	2010-2015 Average	2020	2030	2016-2030 Average
External debt (nominal) 1/	**62.8**	**61.5**	**62.0**			**72.6**	**78.0**	**78.6**	**75.5**	**71.6**	**67.9**		**56.3**	**42.5**	
o/w public and publicly guaranteed (PPG)	46.5	47.3	47.3			61.0	67.3	69.1	66.8	63.7	60.6		49.9	33.5	
Change in external debt	-1.2	-1.2	0.5			10.6	5.3	0.6	-3.1	-3.9	-3.7		-2.0	-2.9	
Identified net debt-creating flows	-10.4	-9.9	1.1			7.8	4.4	1.7	-1.9	-3.9	-1.6		-0.4	-1.6	
Non-interest current account deficit	**12.6**	**10.5**	**8.4**	**7.8**	**4.0**	**16.0**	**14.7**	**13.0**	**10.3**	**7.6**	**7.8**	**11.6**	**7.0**	**2.7**	**5.0**
Deficit in balance of goods and services	34.9	30.2	28.3			35.6	32.4	28.9	25.4	21.6	20.7		18.8	12.5	
Exports	42.8	45.7	36.8			36.1	37.3	39.2	41.1	43.7	45.6	40.5	55.8	57.9	**56.4**
Imports	77.8	75.9	65.1			71.7	69.7	68.1	66.5	65.4	66.3	68.0	74.6	70.4	**71.9**
Net current transfers (negative = inflow)	-22.6	-20.5	-21.8	**-24.7**	**2.7**	-20.9	-18.8	-17.4	-16.6	-15.6	-14.4	-17.3	-13.3	-10.5	**-11.7**
o/w official	-5.1	-6.7	-7.5			-7.1	-5.0	-4.6	-4.0	-3.2	-3.0		-2.3	-0.9	
Other current account flows (negative = net inflow)	0.2	0.8	1.9			1.3	1.1	1.5	1.5	1.5	1.5		1.5	0.7	
Net FDI (negative = inflow)	**-14.3**	**-13.5**	**-7.5**	**-8.1**	**3.7**	**-7.0**	**-8.4**	**-8.7**	**-8.9**	**-8.8**	**-7.3**	**-8.2**	**-6.5**	**-3.7**	**-4.5**
Endogenous debt dynamics 2/	-8.7	-6.9	0.2			-1.2	-1.8	-2.6	-3.3	-2.7	-2.2		-0.8	-0.5	
Contribution from nominal interest rate	2.0	2.2	1.4			2.1	2.2	2.2	2.1	2.1	2.0		1.9	1.6	
Contribution from real GDP growth	-4.6	-3.3	-2.1			-3.3	-4.0	-4.8	-5.5	-4.7	-4.1		-2.7	-2.2	
Contribution from price and exchange rate changes	-6.2	-5.7	0.9			…	…	…	…	…	…		…	…	
Residual (3-4) 3/	9.2	8.6	-0.6			2.8	0.9	-1.1	-1.1	0.0	-2.1		-1.7	-1.3	
o/w exceptional financing	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt 4/	...	...	47.1			55.4	57.9	58.1	56.4	55.1	53.3		48.2	39.2	
In percent of exports	...	...	128.0			153.4	155.3	148.0	137.2	125.9	117.0		86.4	67.7	
PV of PPG external debt	...	...	32.4			43.7	47.2	48.6	47.7	47.1	46.0		41.9	30.2	
In percent of exports	...	...	88.0			121.2	126.7	123.8	116.0	107.7	101.0		75.0	52.1	
In percent of government revenues	...	...	137.5			189.6	200.1	205.2	203.9	191.7	182.5		158.1	110.1	
Debt service-to-exports ratio (in percent)	8.9	8.5	7.9			9.9	9.4	8.8	7.9	7.7	7.6		7.9	7.6	
PPG debt service-to-exports ratio (in percent)	5.1	4.6	4.2			5.0	4.7	4.6	4.0	4.2	4.4		5.6	6.0	
PPG debt service-to-revenue ratio (in percent)	7.9	7.5	6.6			7.8	7.5	7.6	7.1	7.5	7.9		11.9	12.7	
Total gross financing need (Billions of U.S. dollars)	0.0	0.0	0.1			0.2	0.2	0.1	0.1	0.0	0.1		0.2	0.3	
Non-interest current account deficit that stabilizes debt ratio	13.8	11.7	7.9			5.4	9.3	12.4	13.4	11.5	11.5		9.0	5.6	
Key macroeconomic assumptions															
Real GDP growth (in percent)	8.6	6.2	3.6	6.3	2.0	5.4	5.9	6.8	7.6	6.9	6.3	6.5	5.0	5.2	5.0
GDP deflator in US dollar terms (change in percent)	10.6	10.0	-1.4	4.4	10.6	-4.1	-0.5	2.3	2.1	2.5	3.0	0.9	3.0	3.1	3.0
Effective interest rate (percent) 5/	3.8	4.0	2.4	4.5	1.2	3.3	3.2	3.1	3.0	3.0	3.0	3.1	3.5	3.9	3.7
Growth of exports of G&S (US dollar terms, in percent)	14.1	24.8	-17.8	17.7	14.2	-1.0	8.8	14.9	15.2	16.5	14.2	11.4	11.0	6.9	10.0
Growth of imports of G&S (US dollar terms, in percent)	28.6	14.1	-12.4	12.2	14.9	11.2	2.5	6.8	7.3	7.7	11.1	7.7	10.1	12.1	8.7
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	32.1	36.4	38.3	35.7	10.6	16.0	28.2	14.5	19.1	14.6
Government revenues (excluding grants, in percent of GDP)	27.6	28.1	23.6			23.1	23.6	23.7	23.4	24.6	25.2		26.5	27.4	26.8
Aid flows (in Billions of US dollars) 7/	0.1	0.1	0.1			0.2	0.2	0.2	0.2	0.1	0.1		0.1	0.1	
o/w Grants	0.1	0.1	0.1			0.1	0.1	0.1	0.1	0.1	0.1		0.1	0.1	
o/w Concessional loans	0.1	0.0	0.0			0.1	0.1	0.1	0.1	0.0	0.0		0.1	0.1	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			11.3	8.9	7.9	5.8	3.7	3.7		2.9	1.5	2.4
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			55.6	56.6	59.7	63.9	50.7	52.5		42.5	35.3	38.5
Memorandum items:															
Nominal GDP (Billions of US dollars)	1.3	1.6	1.6			1.6	1.7	1.8	2.0	2.2	2.4		3.6	7.9	
Nominal dollar GDP growth	20.2	16.9	2.1			1.0	5.3	9.2	9.9	9.5	9.5	7.4	8.2	8.4	8.2
PV of PPG external debt (in Billions of US dollars)			0.5			0.7	0.8	0.9	1.0	1.0	1.1		1.5	2.4	
(PVt-PVt-1)/GDPt-1 (in percent)						8.9	7.2	5.9	3.9	3.9	3.3	5.5	2.5	0.4	2.2

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.
2/ Derived as $[r - g - \rho(1+g)]/(1+g+\rho+g\rho)$ times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.
3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.
4/ Assumes that PV of private sector debt is equivalent to its face value.
5/ Current-year interest payments divided by previous period debt stock.
6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.
7/ Defined as grants, concessional loans, and debt relief.
8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 2.Cape Verde: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2010-2030
(In percent)

	Projections							
	2010	2011	2012	2013	2014	2015	**2020**	2030
PV of debt-to GDP ratio								
Baseline	43.7	47.2	48.6	47.7	47.1	46.0	**41.9**	30.2
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2010-2030 1/	43.7	38.5	35.3	34.1	36.2	36.1	**30.6**	16.1
A2. New public sector loans on less favorable terms in 2010-2030 2	43.7	49.8	53.3	53.3	52.3	51.9	**50.2**	43.5
A3. Low growth Scenario 7/	43.8	48.0	49.9	49.7	49.7	49.3	46.4	36.0
A4. High Growth Scenario 7/	43.8	46.7	47.4	46.0	44.9	43.1	37.8	24.8
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2011-2012	43.7	47.8	50.5	49.6	49.0	47.9	**43.5**	31.4
B2. Export value growth at historical average minus one standard deviation in 2011-2012 3/	43.7	49.0	55.9	54.4	52.2	49.7	**42.0**	30.2
B3. US dollar GDP deflator at historical average minus one standard deviation in 2011-2012	43.7	50.0	56.2	55.2	54.5	53.3	**48.4**	34.9
B4. Net non-debt creating flows at historical average minus one standard deviation in 2011-2012 4/	43.7	47.9	49.0	48.0	47.3	46.2	**41.9**	30.2
B5. Combination of B1-B4 using one-half standard deviation shocks	43.7	44.8	47.6	47.3	47.5	47.1	**44.2**	31.9
B6. One-time 30 percent nominal depreciation relative to the baseline in 2011 5/	43.7	67.5	69.5	68.3	67.4	65.9	**59.9**	43.2
PV of debt-to-exports ratio								
Baseline	121.2	126.7	123.8	116.0	107.7	101.0	**75.0**	52.1
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2010-2030 1/	121.2	103.2	89.9	83.0	82.8	79.1	**54.9**	27.8
A2. New public sector loans on less favorable terms in 2010-2030 2	121.2	133.6	135.9	129.6	119.7	113.9	**89.9**	75.1
A3. Low growth Scenario 7/	121.5	127.7	125.5	118.4	110.5	104.5	80.1	59.9
A4. High Growth Scenario 7/	121.5	126.1	122.5	114.1	105.4	97.9	70.1	44.4
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2011-2012	121.2	126.4	123.8	116.0	107.7	101.0	**75.0**	52.1
B2. Export value growth at historical average minus one standard deviation in 2011-2012 3/	121.2	138.1	166.4	154.4	139.4	127.3	**87.9**	60.8
B3. US dollar GDP deflator at historical average minus one standard deviation in 2011-2012	121.2	126.4	123.8	116.0	107.7	101.0	**75.0**	52.1
B4. Net non-debt creating flows at historical average minus one standard deviation in 2011-2012 4/	121.2	128.4	124.9	116.7	108.2	101.2	**75.0**	52.1
B5. Combination of B1-B4 using one-half standard deviation shocks	121.2	117.2	117.2	111.3	105.0	99.8	**76.6**	53.3
B6. One-time 30 percent nominal depreciation relative to the baseline in 2011 5/	121.2	126.4	123.8	116.0	107.7	101.0	**75.0**	52.1
PV of debt-to-revenue ratio								
Baseline	189.6	200.1	205.2	203.9	191.7	182.5	**158.1**	110.1
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2010-2030 1/	189.6	163.0	149.0	145.9	147.4	143.0	**115.8**	58.8
A2. New public sector loans on less favorable terms in 2010-2030 2	189.6	211.0	225.2	227.7	213.0	205.8	**189.7**	158.6
A3. Low growth Scenario 7/	190.0	203.5	209.5	209.2	197.2	190.1	170.4	127.7
A4. High Growth Scenario 7/	190.0	197.4	201.6	199.6	187.1	175.8	146.5	93.0
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2011-2012	189.6	202.6	213.3	212.0	199.4	189.8	**164.4**	114.5
B2. Export value growth at historical average minus one standard deviation in 2011-2012 3/	189.6	207.3	236.3	232.4	212.5	197.1	**158.8**	110.1
B3. US dollar GDP deflator at historical average minus one standard deviation in 2011-2012	189.6	211.8	237.4	236.0	221.9	211.2	**183.0**	127.4
B4. Net non-debt creating flows at historical average minus one standard deviation in 2011-2012 4/	189.6	202.7	207.0	205.1	192.5	183.0	**158.2**	110.1
B5. Combination of B1-B4 using one-half standard deviation shocks	189.6	189.9	200.9	202.3	193.3	186.6	**167.0**	116.4
B6. One-time 30 percent nominal depreciation relative to the baseline in 2011 5/	189.6	285.7	293.7	291.9	274.5	261.3	**226.4**	157.6

Table 2.Cape Verde: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2010-2030 (continued)
(In percent)

	Projections							
	2010	2011	2012	2013	2014	2015	**2020**	2030
Debt service-to-exports ratio								
Baseline	5.0	4.7	4.6	4.0	4.2	4.4	**5.6**	6.0
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2010-2030 1/	5.0	4.5	3.7	0.4	-0.7	-0.5	**3.4**	3.2
A2. New public sector loans on less favorable terms in 2010-2030 2	5.0	4.7	4.9	4.7	4.5	5.3	**6.5**	8.7
A3. Low growth Scenario 7/	5.0	4.8	4.6	4.0	4.3	4.4	6.0	6.7
A4. High Growth Scenario 7/	5.0	4.7	4.6	3.9	4.1	4.2	5.4	5.4
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2011-2012	5.0	4.7	4.6	4.0	4.2	4.4	**5.6**	6.0
B2. Export value growth at historical average minus one standard deviation in 2011-2012 3/	5.0	5.0	5.6	6.3	8.9	8.6	**6.8**	7.0
B3. US dollar GDP deflator at historical average minus one standard deviation in 2011-2012	5.0	4.7	4.6	4.0	4.2	4.4	**5.6**	6.0
B4. Net non-debt creating flows at historical average minus one standard deviation in 2011-2012 4/	5.0	4.7	4.7	4.4	4.4	4.6	**5.6**	6.0
B5. Combination of B1-B4 using one-half standard deviation shocks	5.0	4.7	4.4	2.7	2.7	3.0	**5.7**	6.1
B6. One-time 30 percent nominal depreciation relative to the baseline in 2011 5/	5.0	4.7	4.6	4.0	4.2	4.4	**5.6**	6.0
Debt service-to-revenue ratio								
Baseline	7.8	7.5	7.6	7.1	7.5	7.9	**11.9**	12.7
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2010-2030 1/	7.8	7.1	6.2	0.6	-1.3	-1.0	**7.2**	6.7
A2. New public sector loans on less favorable terms in 2010-2030 2	7.8	7.5	8.1	8.3	7.9	9.5	**13.7**	18.3
A3. Low growth Scenario 7/	7.8	7.6	7.7	7.1	7.6	8.1	12.7	14.3
A4. High Growth Scenario 7/	7.8	7.4	7.5	6.8	7.3	7.6	11.3	11.2
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2011-2012	7.8	7.6	7.9	7.4	7.8	8.3	**12.4**	13.2
B2. Export value growth at historical average minus one standard deviation in 2011-2012 3/	7.8	7.5	7.9	9.5	13.5	13.3	**12.3**	12.7
B3. US dollar GDP deflator at historical average minus one standard deviation in 2011-2012	7.8	7.9	8.8	8.2	8.7	9.2	**13.8**	14.6
B4. Net non-debt creating flows at historical average minus one standard deviation in 2011-2012 4/	7.8	7.5	7.7	7.7	7.9	8.2	**11.9**	12.7
B5. Combination of B1-B4 using one-half standard deviation shocks	7.8	7.5	7.6	4.8	4.9	5.7	**12.4**	13.4
B6. One-time 30 percent nominal depreciation relative to the baseline in 2011 5/	7.8	10.7	10.9	10.1	10.8	11.4	**17.0**	18.1
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	0.0	0.0	0.0	0.0	0.0	0.0	**0.0**	0.0

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.

3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).

4/ Includes official and private transfers and FDI.

5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.

6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.
7/ Assumes growth rates respectively 1 percentage points above and below the baseline in 2011-16.

Table 3.Cape Verde: Public Sector Debt Sustainability Framework, Baseline Scenario, 2007-2030
(In percent of GDP, unless otherwise indicated)

	Actual					Estimate	Projections									
	2007	2008	2009	Average	Standard Deviation	2010	2011	2012	2013	2014	2015	2010-15 Average	2020	2030	2016-30 Average
Public sector debt 1/	68.0	65.0	63.9			80.8	86.5	87.6	84.1	79.4	74.5		60.5	41.3	
o/w foreign-currency denominated	46.5	47.3	47.3			62.8	67.5	69.1	66.8	63.7	60.6		49.9	33.5	
Change in public sector debt	-11.6	-3.1	-1.1			16.9	5.7	1.1	-3.5	-4.7	-4.9		-2.1	-2.7	
Identified debt-creating flows	-12.6	-2.1	-0.8			14.9	4.4	0.0	-4.4	-4.4	-4.6		-2.0	-2.8	
Primary deficit	**-1.0**	**-0.3**	**5.3**	**4.2**	**5.1**	**11.8**	**9.6**	**6.4**	**2.5**	**2.0**	**1.3**	**5.6**	**1.8**	**-0.3**	**1.3**
Revenue and grants	32.8	33.6	29.0	**30.3**	**2.5**	30.1	28.6	28.3	27.4	27.8	28.3	**28.4**	28.7	28.2	**28.6**
of which: grants	5.2	5.5	5.4			7.1	5.0	4.6	4.0	3.2	3.0		2.2	0.8	
Primary (noninterest) expenditure	31.8	33.3	34.3	**34.5**	**3.9**	42.0	38.2	34.7	29.9	29.8	29.6	34.0	30.5	27.9	**29.9**
Automatic debt dynamics	-10.3	-1.1	-6.2			3.2	-5.0	-6.4	-7.1	-6.4	-6.0		-3.8	-2.5	
Contribution from interest rate/growth differential	-6.5	-4.0	-1.9			-2.9	-4.4	-5.8	-6.7	-6.0	-5.4		-3.3	-2.1	
of which: contribution from average real interest rate	-0.2	0.0	0.3			0.4	0.0	-0.3	-0.5	-0.6	-0.7		-0.3	0.1	
of which: contribution from real GDP growth	-6.3	-4.0	-2.2			-3.3	-4.5	-5.5	-6.2	-5.4	-4.7		-3.0	-2.2	
Contribution from real exchange rate depreciation	-3.8	2.9	-4.3			6.1	-0.6	-0.6	-0.3	-0.4	-0.6		...	...	
Other identified debt-creating flows	-1.4	-0.7	0.1			-0.1	-0.1	0.0	0.1	0.1	0.1		0.0	0.0	
Privatization receipts (negative)	-1.4	-0.7	0.1			-0.1	-0.1	0.0	0.1	0.1	0.1		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	1.0	-0.9	-0.3			2.0	1.3	1.1	0.9	-0.3	-0.3		-0.1	0.1	
Other Sustainability Indicators															
PV of public sector debt	**21.5**	**17.7**	**48.9**			**61.7**	**66.3**	**67.1**	**65.0**	**62.8**	**59.9**		**52.5**	**38.0**	
o/w foreign-currency denominated	0.0	0.0	32.4			43.7	47.2	48.6	47.7	47.1	46.0		41.9	30.2	
o/w external	...	...	32.4			43.7	47.2	48.6	47.7	47.1	46.0		41.9	30.2	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	4.6	4.7	11.1			17.2	14.8	11.8	7.9	8.7	8.2		9.0	8.5	
PV of public sector debt-to-revenue and grants ratio (in percent)	65.6	52.6	168.7			204.9	231.9	237.4	237.5	226.0	211.9		182.9	134.6	
PV of public sector debt-to-revenue ratio (in percent)	78.0	62.9	207.6			267.6	280.9	283.5	277.9	255.7	237.4		198.3	138.7	
o/w external 3/	…	…	137.5			189.6	200.1	205.2	203.9	191.7	182.5		158.1	110.1	
Debt service-to-revenue and grants ratio (in percent) 4/	17.0	14.8	20.2			17.9	18.4	19.1	19.8	24.1	24.1		25.2	31.2	
Debt service-to-revenue ratio (in percent) 4/	20.3	17.7	24.8			23.4	22.3	22.8	23.2	27.3	27.0		27.4	32.1	
Primary deficit that stabilizes the debt-to-GDP ratio	10.6	2.8	6.4	**6.6**	**3.9**	-5.1	3.8	5.3	6.0	6.6	6.2	**3.8**	3.9	2.4	**3.5**
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	8.6	6.2	3.6	6.3	2.0	5.4	5.9	6.8	7.6	6.9	6.3	6.5	5.0	5.2	5.0
Average nominal interest rate on forex debt (in percent)	1.1	1.1	0.0	1.2	0.5	0.7	0.8	0.9	0.9	0.9	0.9	0.9	1.3	2.2	1.6
Average real interest rate on domestic debt (in percent)	2.5	1.9	0.6	4.0	2.8	2.7	1.7	0.2	-0.3	-0.6	-0.6	0.5	-0.5	-0.6	-0.6
Real exchange rate depreciation (in percent, + indicates depreciation)	-8.0	6.7	-9.4	-1.8	12.8	13.5	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	1.4	2.5	4.2	1.2	2.3	3.3	2.8	2.8	2.6	3.0	3.0	2.9	3.0	3.1	3.0
Growth of real primary spending (deflated by GDP deflator, in percent)	0.0	0.1	0.1	0.1	0.2	0.3	0.0	0.0	-0.1	0.1	0.1	0.0	0.1	0.0	0.0
Grant element of new external borrowing (in percent)	...	...	...	…	…	32.1	36.4	38.3	35.7	10.6	16.0	28.2	14.5	19.1	...

Sources: Country authorities; and staff estimates and projections.

1/ [Indicate coverage of public sector, e.g., general government or nonfinancial public sector. Also whether net or gross debt is used.]

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

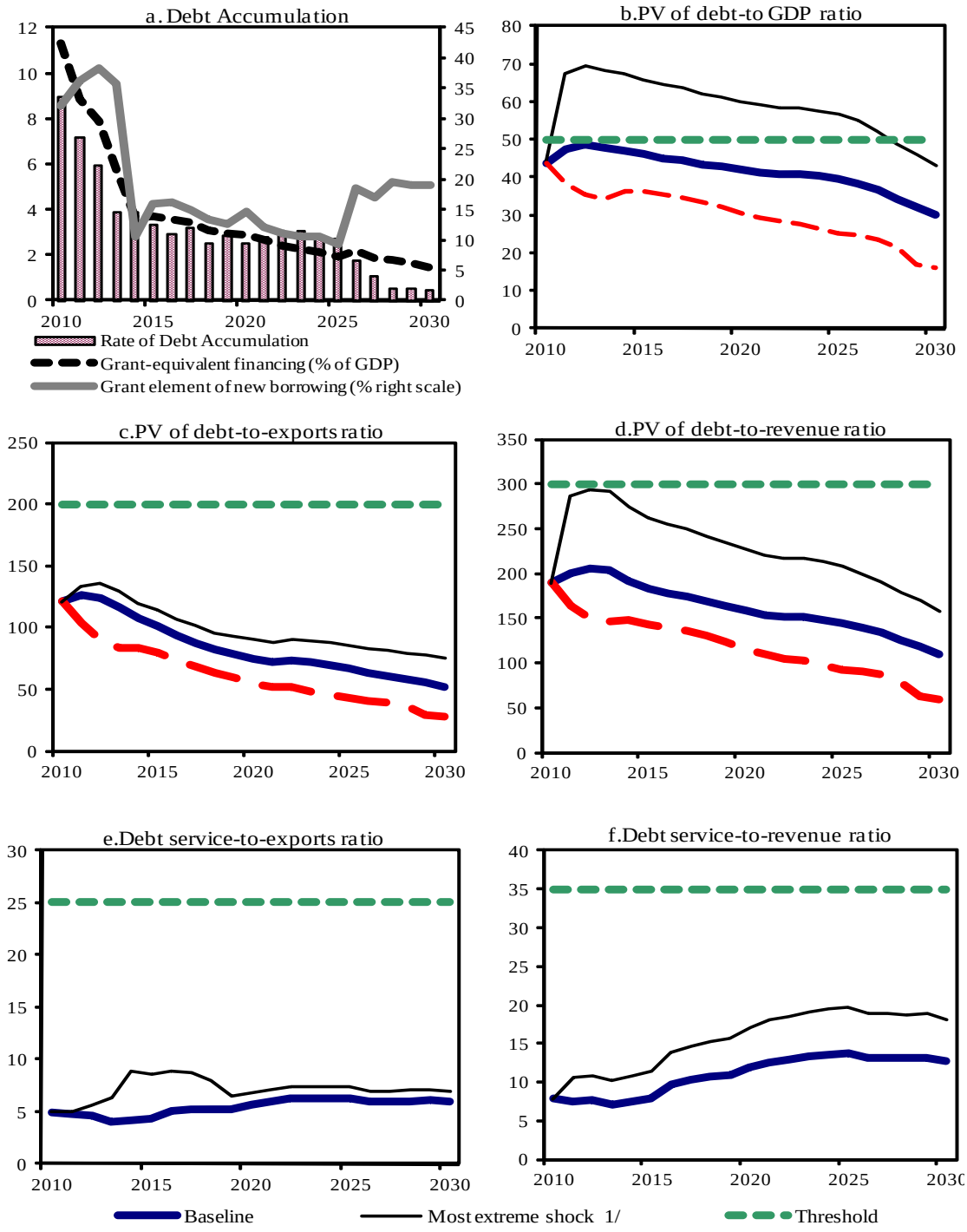
5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 4.Cape Verde: Sensitivity Analysis for Key Indicators of Public Debt 2010-2030

	Projections							
	2010	2011	2012	2013	2014	2015	2020	2030
PV of Debt-to-GDP Ratio								
Baseline	62	66	67	65	63	60	52	38
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	62	61	60	61	61	62	64	73
A2. Primary balance is unchanged from 2010	62	69	76	83	91	98	138	214
A3. Permanently lower GDP growth 1/	62	67	68	67	66	64	62	67
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2011-2012	62	68	72	70	69	67	65	60
B2. Primary balance is at historical average minus one standard deviations in 2011-2012	62	66	70	68	66	63	56	42
B3. Combination of B1-B2 using one half standard deviation shocks	62	64	66	65	63	61	57	48
B4. One-time 30 percent real depreciation in 2011	62	83	82	79	77	73	66	55
B5. 10 percent of GDP increase in other debt-creating flows in 2011	62	77	78	75	73	70	62	47
PV of Debt-to-Revenue Ratio 2/								
Baseline	205	232	237	238	226	212	183	135
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	205	212	211	221	221	218	224	259
A2. Primary balance is unchanged from 2010	205	241	268	304	327	348	482	757
A3. Permanently lower GDP growth 1/	205	234	242	244	235	224	215	236
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2011-2012	205	237	252	256	248	237	225	211
B2. Primary balance is at historical average minus one standard deviations in 2011-2012	205	232	249	250	238	224	195	149
B3. Combination of B1-B2 using one half standard deviation shocks	205	223	233	236	227	215	197	169
B4. One-time 30 percent real depreciation in 2011	205	291	291	289	276	260	229	196
B5. 10 percent of GDP increase in other debt-creating flows in 2011	205	270	275	275	262	247	216	167
Debt Service-to-Revenue Ratio 2/								
Baseline	18	18	19	20	24	24	25	31
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	18	18	18	19	19	17	30	53
A2. Primary balance is unchanged from 2010	18	18	20	21	30	36	81	166
A3. Permanently lower GDP growth 1/	18	18	19	20	25	25	29	50
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2011-2012	18	19	20	21	26	27	32	47
B2. Primary balance is at historical average minus one standard deviations in 2011-2012	18	18	19	20	24	27	28	33
B3. Combination of B1-B2 using one half standard deviation shocks	18	19	19	20	22	22	26	37
B4. One-time 30 percent real depreciation in 2011	18	20	22	23	28	29	36	54
B5. 10 percent of GDP increase in other debt-creating flows in 2011	18	18	21	22	35	35	33	37

Sources: Country authorities; and staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.
2/ Revenues are defined inclusive of grants.

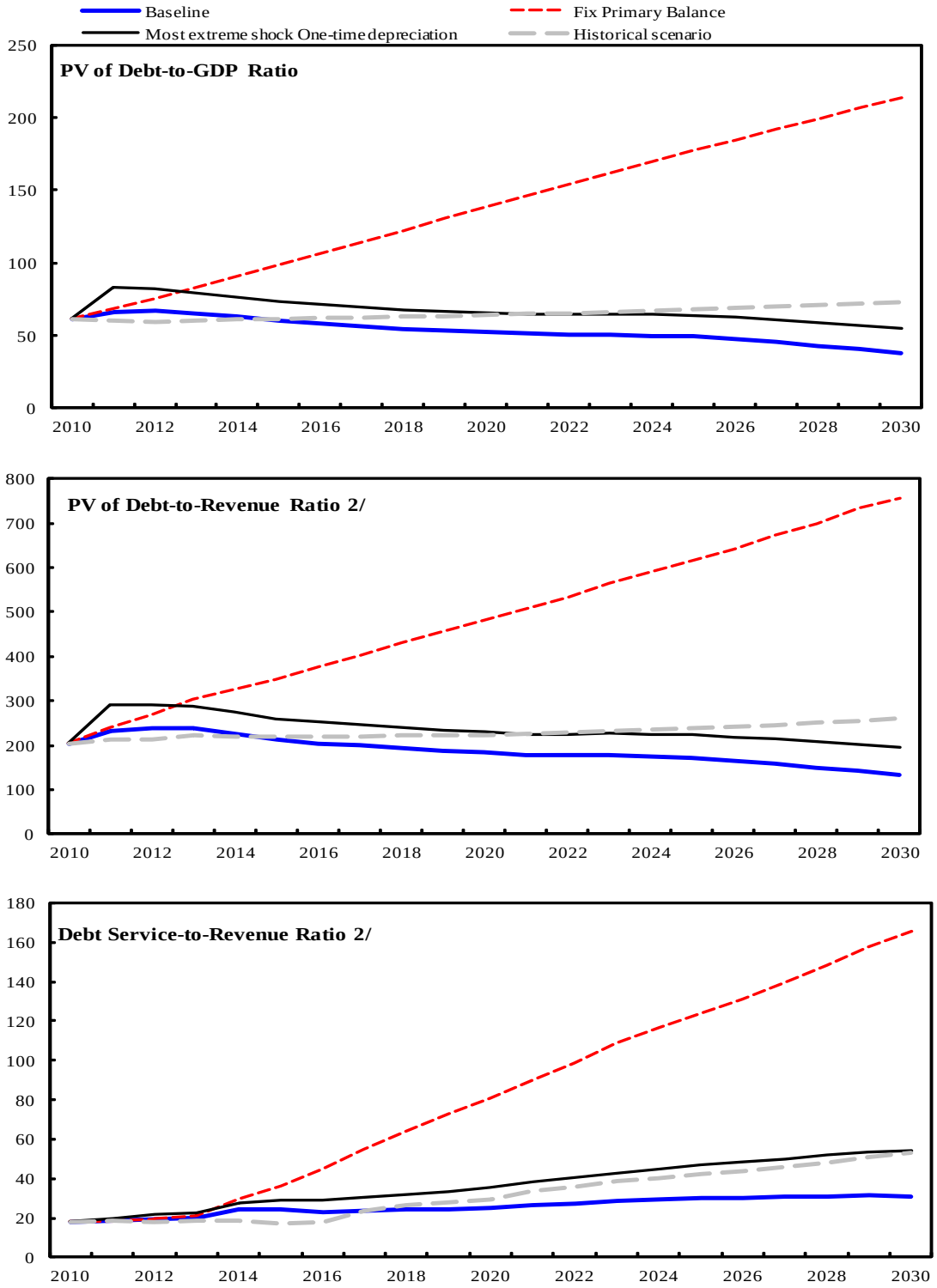
Figure 1. Cape Verde: Indicators of Public and Publicly Guaranteed External Debt under Alternatives Scenarios, 2010-2030 1/



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in 2020. In figure b. it corresponds to a One-time depreciation shock; in c. to a Terms shock; in d. to a One-time depreciation shock; in e. to a Exports shock and in figure f. to a One-time depreciation shock

Figure 2.Cape Verde: Indicators of Public Debt Under Alternative Scenarios, 2010-2030 1/



Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio in 2020.
2/ Revenues are defined inclusive of grants.